PLASTICITY OF THE DIGITAL IMAGE AND SYNTHETIC REALISM IN CONTEMPORARY HOLLYWOOD CINEMA WITH AN EDITORIAL APPROACH TO VISUAL EFFECTS WORKFLOW

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Abstract

This thesis explores the transformation of filmmaking due to computerization and the shift to digital, specifically focusing on the implementation of digital visual effects. The thesis examines the field of visual effects from different perspectives, including their impact on the fundamental component, the image, their narrative integration, and the aesthetic and theoretical implications with special regard to cinematic realism theories. The theory of remediation, immediacy, and hypermediacy is applied to demonstrate how digital visual effects can be used to achieve different goals. Furthermore, the thesis examines the intricate relationship between editing and visual effects in theoretical and practical terms, as they both are key techniques of manipulation and creation of meaning, revealing the similarities and differences between montage and composite techniques. The thesis further argues that VFX can be used as a tool for editors, which is an exciting development in the evolution of cinema.

Keywords*: digital cinema; visual effects; indexicality; realism; montage; composite; postproduction; editing*

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I. Introduction

The shift to digital technology in filmmaking has transformed the landscape of cinema since the early 1990s. With the help of complex digital visual effects, filmmakers can bring their vision to life and create compelling narratives in a previously impossible way. However, the new aesthetic of digital cinema has challenged classical film theories, resulting in a debate among scholars and filmmakers alike.

The use of tricks and special effects dates back to the early years of cinema, but as technology changed, they became more sophisticated, realistic and expressive. Furthermore, the democratisation of technology led to the wide range of use of visual effects not only in mainstream and blockbuster Hollywood productions but in independent and European films as well. Incorporating the new practices in narrative and technical terms can be challenging. Therefore, I find it relevant to research this field as an editor because the demand for understanding and adapting to the increasingly complex workflow is growing. “We encourage other independent filmmakers to understand this language because this is how all movies are gonna be made now; there always be some elements of visual effects, even when you can’t tell because the technology has become so easy and effortless to use” (WIRED, 2022) - suggests Daniel Kwan, director of *Everything Everywhere All at Once (2022)* winning 7 Oscars in 2023 including best picture. The movie became famous for its incredibly small, five-person VFX team achieving 500 VFX shots “in their bedroom” (WIRED, 2022).

My motivation behind this research stems from previous difficulties I’ve faced while working with visual effects as an editor. Through this research, I hope to understand better the growing trend of visual effects and how to integrate them effectively into a film. The research will adopt both theoretical and practical approaches. The theoretical framework will draw on film theory to revisit the concept of realism and indexicality and its application to the aesthetics of digital cinema and visual effects. In addition, an interview with Joe Walker, a renowned film editor with extensive experience in major motion pictures, will be conducted to gain insights into the evolving role of editors in the ever-changing workflows.

The field of visual effects is vast and complex. This thesis focuses on a micro-level analysis of the image, specifically examining the ontological change between the celluloid-based and the digital, synthesised images. The goal is to conduct theoretical research and analysis on the implications of visual effects on cinematic realism and to explore the relationship between montage and composite. With the shift of special effects from production to post-production, the boundary between departments like cinematography and editing and visual effects are becoming increasingly blurred, and the role of the editor is evolving by adapting to new practices. This thesis investigates these changes and how visual effects extend the choices made during editing. However, this paper is not intended to provide an in-depth analysis of the current technology of visual effects, nor does it address labour or economic issues within the industry, and it will not investigate the cultural and social implications of mainstream Hollywood cinema.

II. Theoretical Framework

Indexicality and the Photographic Image

In exploring visual effects and digital cinema, it is crucial to examine its fundamental component, the image and shot, and theories about the ontology of the photographic realism.

André Bazin (1967) believed that the camera had the unique ability to record reality objectively and truthfully, capturing the essence of the world through the lens. In his view, cinema is the only art that achieved a faithful representation of reality; unlike a painting, it is free from subjective interpretation due to the mechanical process that “satisfy, once and for all […] our obsession with realism” (Bazin, 1967, p. 12). “All the arts are; based on the presence of man, only photography derives an advantage from his absence” (Ibid, p. 13). Although a realistic painting may appear to resemble the subject more than a monochrome, faded, or defocused photograph, the latter still has a fundamental connection and faithfulness to the original subject. It possesses an irrational power and shares a common existence with the model, much like a fingerprint (Ibid). Cinema, therefore, for Bazin, becomes “objectivity in time” (Ibid, p.14).

Roland Barthes (1981) held a similar perspective on the essence of the photographic image, which is physically connected to its referent as if they were “glued together, limb by limb” (pp. 4-6). He argues that while a drawing is a coded message that requires rules and differentiates between significant and insignificant elements, it cannot reproduce every detail. In contrast, a photograph is a *message without a code* and cannot intervene within the object except through trick effects. The absence of a code by its mechanical means makes the photograph more objective than any other medium (Barthes, 1977).

Despite not using the term *index,* both Bazin’s (1967) and Barthes' (1981) observations about photography closely align with C.S. Pierce's (as cited in Wollen, 2013) concept of an index. According to Pierce’s classification, a sign is either an icon, a symbol, or an index. An *index* is a sign connected to its object through a direct physical relationship that proves its existence. An icon is a sign that primarily represents its object through similarity or likeness. Unlike indexes or icons, a *symbol* requires no existential connection or resemblance to its object. Instead, it is a convention established by agreement or law and exists in the minds of those who use it (Wollen, 2013).

A photograph is indexical because of the direct physical connection between the object and the representation. The image is created by the reflection of light off the object, filtered through the camera's lens and iris to create the photographic negative (Gunning, 2020). Being an index does not require resemblance, only the existential bond between object and representation. Therefore, in photography, iconicity and indexicality is usually present together, as the photograph resembles the object it represents and has the evidential and physical connection to its referent, a sign of its existence, like a fingerprint (Film & Media Studies, 2021a).

Realist and Formalist Approach

Classical film theory aims to define and identify what cinema is and what makes it different or legitimates it as an art form. Bazin (1967) distinguished between “directors who put their faith in the image and those who put their faith in reality” (p. 24). By the first category, he meant “everything that the representation on the screen adds to the object”, mainly through montage and the plasticity of the image (Ibid, p. 24). Bazin (1967), as a prominent realist theoretician, believed that the essence of the film lies in its capability to capture reality in a truthful way, and “the image is evaluated not according to what it adds to reality but what it reveals of it” (p. 28). To permit the reality of the image to be the driving force behind the cinematic content rather than forcefully manipulating the medium to draw the viewer's attention to them (Berton, 1990).

On the other hand, Arnheim (1957), as a leading formalist thinker, argues with the realist concepts that lens-based art's only function is the mere reproduction of reality. In his view, art begins when the mode of presentation adds to the object. According to Arnheim's *partial illusion* theory, film can create only the illusion, a two-dimensional representation of reality by its technical limitations and unique characteristics. These include the chosen perspective and framing, the spectator's position and viewpoint, the use of editing that interrupts the continuity of space and time, artificial lighting, colour, sound, and camera movements (Ibid). In his perspective, the camera is not simply a tool for recording reality but rather an interpreter that actively contributes to the creative process. The camera's artificiality and the filtered representation are essential components of artistic expression (Berton, 1990). Therefore, digital cinema has even more possibilities because of the complex and more controllable filtering process (Ibid).

Realism in cinema aims to achieve a realistic design by limiting the stylistic choices that might distort the portrayal of characters or places. The goal is to create a representation that is true to life without the embellishments of style (Realism and Formalism, 2020). Bazin (1967) and the realist aesthetic values also the limitation of editing, seeing it as means of manipulation and a tool for directing focus.

The use of visual effects moves away from realism towards highly stylised images that can deviate from the reality captured by the camera, based on manipulating the image rather than capturing reality. Therefore, it belongs to the domain of the illusory and the formative tendency in filmmaking (Prince, 2012).

Cinema combines elements of both realism and formalism to create a unique and captivating experience (Lewis et al., 1975). Perkins (as cited in Lewis et al., 1975) tries to establish a more rational aesthetic, saying that none of the approaches are better; both practices are cinematic and have different functions that must be matched to certain types of films. Similarly, Prince (1996) suggests that cinema cannot be divided by either indexical recordings or stylistic transfiguration of the world because it does both. Realism and formalism are general terms; however, we should rather consider them as two poles of a spectrum than absolute qualities. Very few films are purely realist or purely formalist; most of them fall somewhere between these two extremes, what film theory refers to as classical Hollywood narration (Realism and Formalism, 2020), which is characterised by its invisibility and the transparency of the medium through seamless continuity editing, functional visual style, and character-driven three-act narrative structure (see Bordwell et al., 1985, for more).

The development of digital technology and visual effect techniques has merged the two opposing forces in filmmaking by enabling the synthetic creation of photorealistic images without any indexical referent. The boundaries between authenticity and artifice are blurred and distinguishing between real and synthesised images is increasingly difficult (Dancyger, 2018). Prince (2012) suggests that to understand visual effects fully, we must move beyond film theory's enduring dichotomy. As photographic indexicality is typically considered the foundation of realism in cinema, visual effects create issues between realism and the indexical relationship.

It is also important to note that aesthetic realism as a film style is different from the ontological realism of the image. Metz (1991) suggests that there needs to be a clear distinction between the two concepts: the impression of reality created by the narrative of a work of fiction and the actual reality of the medium representing that narrative. His argument is to consider both aspects separately, and he points out that using the term ‘real’ can be confusing and can lead to misunderstandings (Ibid). In contemporary film theory, *cinematic realism* aims for transparency, and the indexicality of *photographic realism* is replaced by the ‘reality effect’ achieved through codes and discourse (Prince, 1996).

Defining Digital Cinema

The digital image is not a photochemical footprint of the world, not analogue, containing endless information, but consists of discrete elements, pixels that refer to memory locations stored in numbers (Rodowick, 2007).

In *The Matrix (1999),* a fictional character, Cypher - unlike any real human – can read the computer's binary codes without any conversion *(Figure 1).* The difference in the interface (digital computer screen versus projection of an analogue image) and the fact of being *coded* – unlike the photographic image, a *message without a code*– indicates the ontological difference between analogue and digital images (Rodowick, 2007).

***Figure 1** Cypher is reading the digital binary code. Source: The Matrix (1999)*

If classical film theory aimed to identify the purpose of cinema as a new medium and its unique features as an art form, contemporary film theory faces a similarly challenging task in attempting to define digital cinema. Rodowick (2007) raised questions such as: "Can information processing be considered a creative medium? Can the computer as a simulation machine or information processor give rise to creative automatisms?" (p. 99).

Manovich (2002) suggests that with the advent of digital technology, there has been a significant change in the film industry, exemplified by the increasing use of computer-generated special effects in Hollywood movies during the 1990s. The shift to digital has impacted not only Hollywood but also the entire filmmaking industry. After summarising the characteristics of digital cinema, and he gives the following definition (Ibid, pp. 254-255):

digital film = live action material + painting + image processing + compositing + 2D computer animation + 3D computer animation

His characterisation will be followed and applied for further analysis:

Losing the Indexicality

"Here we confront a new kind of ontological perplexity – how to place or situate ourselves in space and in time in relation to an image that does not seem to be "one."
(Rodowick, 2007, p. 94).

In the past, despite genre or style, a common feature of all movies was their "reliance on lens-based recordings of reality," cinema was defined as "the art of the index" (Manovich, 2002, pp. 249-250). However, the computer treats all digital images the same way, as collections of easily modified pixels, regardless of whether they were recorded through the lens or synthesised in 3D graphics (Ibid). The *inherent mutability* is a key feature of the digital image; it is easy and quick to manipulate, to replace old digits with new ones. The digital artist's tools are similar to the painter's brush, meaning, that it "erases the difference between a photograph and a painting" (Ibid, p. 256). The absence of a direct link between computer-generated imagery (CGI) and reality challenges the earlier presented Bazinian realism theory based on the ontology of the image (Ibid).

Pierce (as cited in Wollen, 2013) argued that indexes do not require any similarity or resemblance to their causes but instead serve as signs of existence. Consequently, the iconicity, the resemblance of the image to the object, does not guarantee indexicality without a physical relationship (Rodowick, 2007). While digital photos can also serve as indexical signs like analogue photos, they have different capabilities and functionalities compared to film. While analogue recording is a form of transcription, digital capturing involves converting a non-quantifiable image into a mathematical notation, resulting in discontinuous data that weakens the indexical link to physical reality. Unlike changes made on an analogue image, alterations to digital data are reversible and can be undone. Similarly, digitalising an analogue image is an irreversible operation (Ibid). Furthermore, in the case of a computer-generated image, light simulation in the computer does not need a physical source, and shadows are created regardless of the position of existing lights. In photography, lighting plays a crucial role in creating the exposure and final image, whereas in digital imaging, it is a matter of painting and adjusting individual pixels' brightness and colour data (Ibid). This means that lighting in computer imagery is not bound by physical conditions that photographs require for their creation. As a result, digital imaging may fail to deliver the idea of indexicality and the being of a fingerprint of the physical world (Ibid). However, photographic truth and indexicality are fundamental in documentary filmmaking, unlike fiction films, where truth is not necessarily as relevant in the narrative context (Prince, 2012).

The use of digital imaging allows for the creation of objects that appear to be anchored in photographic reality through realistic lighting, shadows, reflections, and surface textures that respond to the touch of live actors. In addition to its ability to produce realistic images, digital imaging can also distort the recording of real physical objects in ways that challenge indexical referentiality (Prince, 1996). For example, in *The Abyss (1989),* a fictional creature, the "pseudopod"[1] was born on computer without any real-life referent to ground its indexicality (Ibid).

Figure 2 *Interaction of a live-action character with a non-indexical computer-generated character. Source: The Abyss (1989)*

We need to note that manipulating the filmic image was also possible in the analogue era, although the different elements of the image were unified in front of the camera lens (Keil & Whissel, 2016). Digital synthesis is a process that generates intentionally fictional images with no basis in reality; they are immaterial. In contrast, double exposures, analogue matte-paintings, and optical printing use elements recorded from physical spaces, combining them to create an imaginative effect from causal elements. Digital compositing and computer-generated imagery can generate a unique outcome with ontological peculiarity (Rodowick, 2007).

[1] the watery creature in the movie illustrated on *Figure 2*

Plasticity and Synthetic Realism

> "The fantastic creatures of King Kong were drawn, but the drawings were then filmed, and that is where for us, the problem begins" (Metz, 1991, p. 5).

Manovich (2002) continues that digital imagery has allowed film to retain visual (photo)realism while having the plasticity of the handcrafted image, and filmmaking returned to the manual construction of the image, "painting in time […] no longer a kino-eye, but a kino-brush" (pp. 259). He further explains that this trend can be seen as a return to the pre-cinematic moving picture techniques, like the magic lantern slides, the Phenakistiscope, or the Zoetrope, with their hand-crafted images. "The history of the moving image thus makes a full circle. Born from animation, cinema pushed animation to its boundary, only to become one particular case of animation in the end" (Ibid, pp. 254-255). This also means that digital image should be considered a medium for expression rather than reproduction (Rodowick, 2007).

Cinema, especially classic Hollywood narration, intentionally tries to hide the construction behind the image, resulting in a new kind of realism, which aims to create the perception that what is shown on the screen is an honest representation of the world "rather than creating the 'never-was' of special effects. […] Something which is intended to look exactly as if it could have happened, although it really could not" (Manovich, 2002, pp. 253-254). To describe the distinct ontology of digital images based on perception rather than the indexicality of photographic images, Prince (1996) uses the term *'perceptual realism'* (Ibid). Computer-generated images are artificially designed to match the viewer's audio-visual perception of reality instead of having a direct connection to the real world. "The resulting image is perceptually realistic but referentially unreal, a paradox" (Ibid, p. 34).

The computer can create the photorealistic impression that an object exists, but the artist's responsibility is to convey to the viewer that the object's existence has meaning (Berton, 1990). Fantasy, action, or sci-fi films – genres that typically use visual effects - follow a different logic from reality, a genre logic. Genres are institutionalised sets of rules that a film adhere to from the outset, and within those rules, it remains perfectly consistent and coherent (Metz & Guzzetti, 1976). A film's "inner truth" is crucial, to remain faithful to its own concept of reality, established rules and principles, rather than actual reality. Any inconsistency can disrupt the story's continuity and distract the audience. In a cinematic work, when verisimilitude is established, viewers agree to believe in the fiction and are willing to suspend their disbelief (Realism and Formalism, 2020).

To achieve ’perceptual realism’, digital effects, objects, or creatures must be anchored in the Newtonian space, or as Whissel (2014) refers to *“staging the indexicality”* (p.115). She offers methods for establishing the digital entities' materiality and embodied presence in the fictional world. For example, the materiality of King Kong or Godzilla was established through their enormous footprints, as their physical traces are present in the fictional world (Ibid). Strategies, such as live-action characters in the movie attempting to capture or photograph *(The Lost World, 1997*; *Godzilla,1998*; *Cloverfield, 2008),* the digital creatures further validate their materiality. In *King Kong (2005),* there is a self-referential line in which a character insists on including themselves in a photo of the creatures to prevent people from claiming ‘it is fake’ (Ibid). To emphasise the mass and weight of the immaterial creatures, movies such as *Jumanji (1995)* and *Jurassic Park (1993)* have used techniques like shaking the ground or showing a shaking cup before the arrival of the creatures. Finally, the interaction between live-action characters and CGI creatures, often involving endangerment or harm to humans, further validates their presence and the consequences of their presence in the fictional world. (Ibid).

Figure 3 *CGI dinosaur in Jurassic Park (1993)*

As discussed in Film and Media Studies (2021b), several of these methods are used and demonstrated the 'staged indexicality' in *Jurassic Park (1993)*, when the first CGI dinosaur in the movie, the Brachiosaurus, arrives *(Figure 3).* Firstly, the dinosaur is intentionally too large to fit the frame, emphasising its massive size. The camera then pans up to reveal it, creating the impression that it was filmed with a physical camera. Secondly, the live-action characters are composited together with the CG creature, imitating a singular point-of-view that mimics the fact of being captured together. Thirdly, to emphasise the dinosaur's physical weight and mass, the ground shakes in the shot when the creature branches and then returns to the ground, as well as in the following shots. This creates the impression that all the characters occupy the same physical space (Film & Media Studies, 2021b).

We might mistakenly think that computer graphics can fake reality because we have grown accustomed to accepting photographs and films as accurate representations of reality over the past century and a half. The result of computer graphics is not the representation of reality, but a representation of reality as seen through the camera lens. Therefore, what is being faked is solely the image produced by films, not reality itself (Manovich, 2002).

When capturing reality through the camera lens, everything can be photographed, resulting in an analogue and uniform reality, without distinguishing between significant or insignificant elements - as Barthes (1977) observed. However, constructing reality through 3D computer graphics involves building it from scratch, resulting in an uneven and incomplete reality with gaps and white spots (Manovich, 2002).

Jurassic Park (1993) is a landmark film because it seamlessly integrated live-action footage and mixed physical puppetry with CGI. To match the imperfection of the recorded footage, the CGI images had to be intentionally degraded. Unlike camera recordings, a synthetic image has unlimited resolution and detail (Ibid). To achieve photorealism, two objectives must be met. The first is to simulate real-life scenes, objects, and environments, which requires solving three separate problems: representing an object's shape, lighting effects on its surface, and modelling movement (Ibid). The second goal involves the simulation of traditional cinematography, including the camera artefacts, such as depth of field, film grain, motion blur, and the limited tonal range (Ibid). All components, including live location footage, blue screen footage of actors, and 3D computer-generated elements, must be aligned in perspective with consistent contrast and colour. The depth of field is simulated by blurring certain elements and sharpening others. Furthermore, virtual camera movements, film grain, or video noise can be added to the final product (Ibid).

According to Rodowick (2007), there is a paradox in perceptual realism's insistence on maintaining photographic realism and the stylistic goal of Hollywood cinema for transparency. Simply put, we are bound by photographic realism while not restricted to the photographic process. Manovich (2002) further observes that typically, we associate computerization with automating processes. However, the case of CGI and certain visual effects, the opposite has occurred. Previously, a camera automatically recorded the information, but now it must be painted frame by frame.

Live Action Is Just an Option

Manovich (2002) suggests that cinema's focus on visual realism through (photographic lens-based) automatic recording was a temporary exception in history and not the only way to function anymore. It is possible to generate and imitate realistic and film-like images and to combine them seamlessly with live-action recordings.

Over the past few years, the relationship between live-action and CGI has further evolved, and it is becoming more common to match the live-action footage to the computer-generated world rather than the other way around as it was previously. Graham Jack[2] (DNEG, 2017) points out that visual effects processes have undergone significant changes with computerisation. Originally special effects were part of the filmmaking process, on set, in camera. With computerisation, it changed, and effects and certain elements were added to the live-action plates[3] later. Today, the process is more proactive, with filmmakers shooting natural elements on set that must be matched and incorporated seamlessly into primarily computer-generated scenes and environments (DNEG, 2017). In conclusion, live-action is no longer the only or even the primary option for cinema. As a result, “live action footage is displaced from its role as the only possible material from which the finished film is constructed." (Manovich, 2002, p. 254).

[2] chief technology officer of DNEG (Double Negative) in 2017
[3] Live action footage intended as an element in a composited visual effect shot. Plates often consist of location or sets for use as backgrounds or other elements as needed. https://www.vpglossary.com/vpglossary/plate/

Production Is Shifting Towards Post-Production

The digital workflow has led to an overlap between the previously distinct domains of cinematography, special/visual effects and editing (Keil & Whissel, 2016). In the past, editing and special effects used to be completely separate operations; the editor's job was to arrange a sequence of images, while special effects experts handled any modifications within an image. However, with the introduction of computers, this division no longer exists. The footage captured on set is no longer considered the final product, and the real work of constructing the effect shots takes place in post-production, involving the editor (Manovich, 2002). This means that tasks like reordering images, combining them, making changes to specific parts, or tweaking individual pixels are all essentially the same operation, both in theory and in practice. “In short, the production becomes just the first stage of post-production” (Ibid, pp. 255). The filming of *Stars Wars - The Phantom Menace (1999)* took 65 days, while the postproduction stretched over two years because 95% of the film (approximately 2,000 shots out of the total 2,200) was finished on a computer (Ibid), which in practical terms makes it similar to an animated film (Rodowick, 2007).

While Manovich’s observation was accurate in 2002 and is still considered as the leading and most widespread visual effects workflow, virtual production seems to reverse this process again. With the emerging technology of virtual LED walls, the CGI environment, which is reactive to the camera movements and perspective, can be projected into the background and recorded live on set, allowing immediate changes and more control for the filmmakers and a more immersive experience for the actors. „It is a revolutionary set of technologies which marries the pre-, principal, and post-production processes - and blurs the line between the digital and physical” (Cheung, 2023). In virtual production, as the effects are applied during pre-production and production, the post-production phase is considerably shorter than in traditional projects (Cheung, 2023). Denis Bachter and Anja Siemens, editors of the Netflix show *1899 (2022),* expressed how much virtual production technology helped them during the editing process compared to the traditional visual effects workflow, which can be challenging with lots of green and blue screens and missing elements. “It is special for us in the editorial to have shots kind of finished. It is easier to get more into the emotion; it is not like that I see the studio, and I see the boom [in the shots], I see everything” (Netflix Production Technology Resources, 2022).

Montage and Composite

Editing and visual effects are intertwined crafts. However, the process and their roles are distinct; both are image transformation and manipulation techniques, and both are considered as part of post-production (Keil & Whissel, 2016). Editing usually aims to stay hidden to maintain continuity and seamlessness. On the other hand, visual effects are more connected to the extraordinary and the spectacular but are capable of producing both fantastical and realistic images (Ibid). Montage or editing is the common technique for organising image sequences, juxtaposing discrete elements in time, one shot after another, to create a sense of continuity and to conceive a meaning (Manovich, 2002). Compositing, on the other hand, involves the layering of elements into a seamless image (Ibid) or to create “a whole that is greater than the sum of its parts[4]” as John Dykstra, VFX supervisor phrases it (Light & Magic, 2022a).

Manovich (2002) analyses the differences and similarities between these two techniques. He distinguishes between temporal montage (editing), “separate realities form consecutive moments in time”, and spatial montage (composite) “, montage within a shot […] separate realities from contingent parts of a single image” (p. 140).

[4] Originally attributed to Aristotle, in Metaphysics (Martin, 2021).

Temporal montage

Manovich (2002) argues that cinema is simulation and “editing, or temporal montage, is the key twentieth-century technology and cinema’s main operation for creating fake realities” (pp. 140-141). Soviet film theory focuses on the analysis and examination of the fundamental component of film, the shot. Kuleshov and Eisenstein believed that shots are arranged in a particular manner known as ‘montage’. Similarly, digital compositors may also need to arrange individual components to construct a single shot (Niu, 2020).

A prominent theory and term associated with Kuleshov is the so-called ‘creative geography’, which shows the manipulating power of editing. The spatial and temporal coherence in cinema is sometimes achieved by faking reality and synthesising distinct elements creating the illusion of one fictional space, but those can be recorded at different times and locations (Manovich, 2002). Therefore even “through montage, film can overcome its indexical nature, presenting a viewer with objects [spaces] which never existed in reality” (pp. 140-141).

The well-known Kuleshov-effect[5] is “utilised to change the meanings of individual shots, or, more precisely, to construct a meaning from separate pieces of pro-filmic reality” (Ibid, p. 140). Bazin (1967) explains that the meaning and interpretation are not objectively present in the images but is created by their juxtaposition, and “it is in the shadow of the image projected by montage onto the field of consciousness of the spectator” (p. 26). Similarly, Eisenstein (1957) writes that “two pieces of any kind, placed together, inevitably combine into a new concept, a new quality, arising out of that juxtaposition” (p. 4), which phenomenon is not exclusively unique to montage but the juxtaposition of any two objects (Eisenstein, 1957).

[5] The Kuleshov effect is the idea that two shots in a sequence are more impactful than a single shot by itself. This effect is a cognitive event that allows viewers to derive meaning from the interaction of two shots in sequence. https://www.nfi.edu/kuleshov-effect/

Eisenstein created his “methods of montage”, exploring the creation of meaning and effect by editing different shots together. He believed that montage is collision or conflict in a dialectic manner (Eisenstein, 1949). However, Soviet montage theory differs from Hollywood continuity editing. The Soviet montage theory prioritises conveying a message and creating meaning through editing rather than used mainly to maintain continuity and seamlessness as in classic Hollywood continuity editing, which, however, also benefits the power of ‘creative geography’. In addition, Manovich (2002) points out that Eisenstein’s montage theory only focuses on one dimension: time.

Spatial montage

Film and video technology was designed to display a single image that would completely fill the screen. This meant that filmmakers had to “work against the technology” to be able to experiment with spatial montage. The computerisation broke the one image/screen logic and created new possibilities for filmmakers (Manovich, 2002). However, the composite technique was present from the early years of cinema through double exposure, rear projection or optical effects, montage within a shot, or spatial montage was less commonly used in the pre-digital era simply because of its difficulty and imperfection (Ibid). Manovich (2002) argues that computer culture changed the dominant practice for the “simulation of non-existent spaces” (p. 145), as it is no more relying on montage as the leading technique. Digital compositing creates a different paradigm and provides an “alternative aesthetics of continuity” in space instead of time (Ibid, p. 136). He sets up two conditions for qualifying composite as a type of montage: the juxtaposition of discrete elements and the establishment of meaning, emotion, and aesthetic effects. Based on these conditions, he argues that "digital filmmakers can create what I will call spatial montage" (Ibid, p. 147), which introduces a new dimension in filmmaking by exploring "the position of the images in space in relation to each other" (Ibid, p. 272). Furthermore, he states that:

> Compositing, in general, can be understood as a counterpart of montage aesthetics. Montage aims to create visual, stylistic, semantic, and emotional dissonance between different elements. In contrast, compositing aims to blend them into a seamless whole, a single gestalt (Ibid, p. 136), […] with their boundaries erased rather than foregrounded (Ibid, p. 145).

Therefore, the concept of "montage" in filmmaking is no longer limited to combining elements in time; with modern technology, any element of an image can be changed or manipulated in terms of its value and position (Rodowick, 2007).

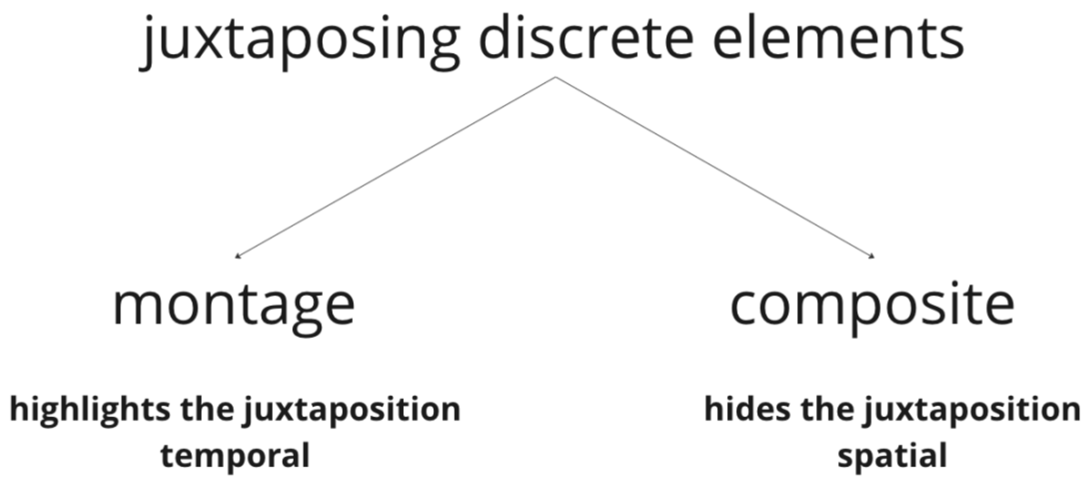

Figure 4 *Temporal and Spatial Montage. Summary of Manovich's theory on the concepts of montage and composite, emphasising the similarities and differences between these operations.*

Remediation

„Along with immediacy and hypermediacy, remediation is one of the three traits of our genealogy of new media" (Bolter & Grusin, 1999, p. 273).

The practice of creatively manipulating photographic images has been around since the invention of photography, and the tension between creating a realistic representation and using artifice to convey a certain message existed long before the emergence of digital imaging (Prince, 1996). "Visual effects are coextensive with narrative film, and digital tools have made them more expressive, persuasive, and immersive" (Prince, 2012, p. 4).

Prince (2012) suggests using the term 'special effects '(SFX)[6] or 'physical effects' for effects created on location during live-action shooting. On the other hand, 'visual effects' (VFX) are digitally created in postproduction after principal photography, creating, altering, or enhancing images that were impossible to achieve during shooting (Ibid).

The theory of remediation by Bolter and Grusin (1999) helps to explain the characteristics and artistic qualities of digital media forms like virtual reality, video games, and CGI films. They described remediation as the tendency for a newer medium to borrow and refurbish techniques from preceding ones, the "representation of one medium in another" (p. 45), while newer media tries to improve and enhance the preceding ones (Ibid).

In the context of visual effects, remediation occurs when digital cinema and computer-generated imagery combine different media and methods, such as painting, computer and motion graphics, and traditional film techniques (Niu, 2020). The influence of

[6] This includes stunt work, makeup effects, miniatures, matte paintings, models, puppetry, and explosions or crashes that are staged during filming and captured by the camera (Prince, 2012). Additionally, 'optical effects' involve using the properties of light, film, and lenses and can include techniques like double exposures and split-screen shots. While the term 'opticals' is often used to describe composites created using the optical printer, 'optical effects' specifically refer to these camera-based processes (Keil & Whissel, 2016).

painting on visual effects can be observed through the involvement of concept artists in the industry. Companies like MPC specifically seek concept artists with a traditional art background (Ibid). Digital visual effects remediate the "physical" special effects techniques. The analogue technique of double exposure, optical tricks, matte painting, rear projection, or optical printing was replaced by compositing or digital matte painting (green screen). The use of miniatures, stop motion puppets or animatronics were replaced by 3D modelling.

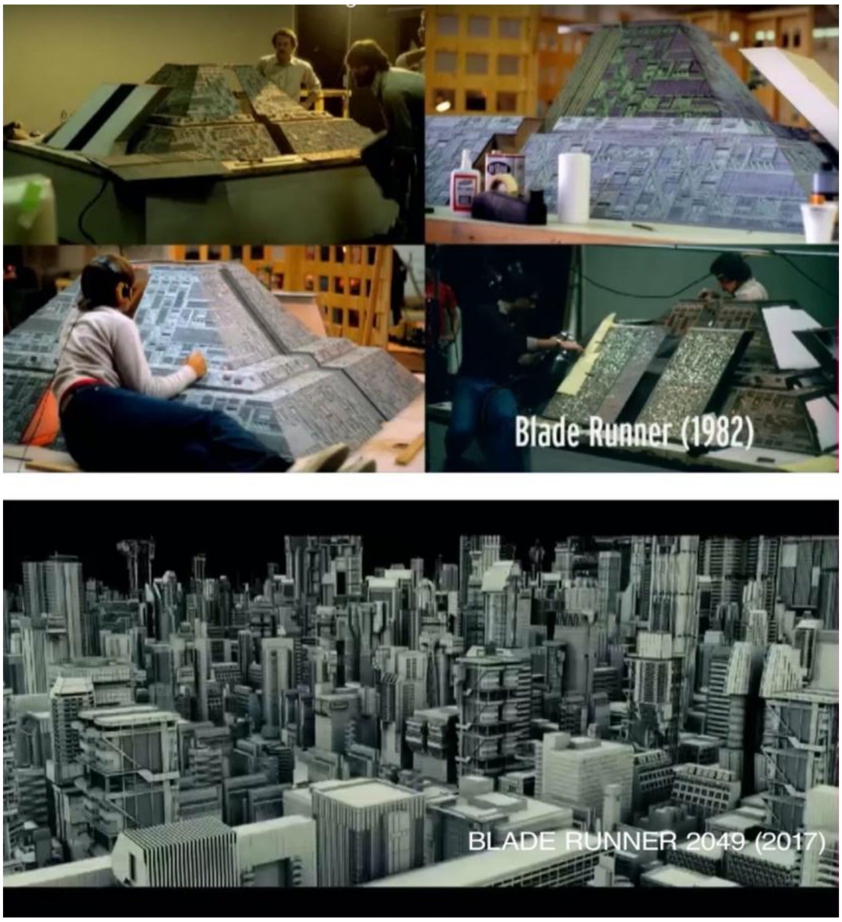

Figure 5 *Miniature technique and 3D modelling. The digital 3D models and environments in Blade Runner 2049 (2017) replaced miniatures used earlier to create cityscapes in Blade Runner (1982). Source: (DNEG, 2017)*

On the 2017 Web Summit, Graham Jack mentions examples of remediation like the pepper's ghost effect[7], traditionally used in theatre or technologies borrowed from the

[7] The Pepper's Ghost effect is a type of holographic projection that utilizes a mirror and glass surface. It has been used in theatre and by early illusionists. Additionally, it has been used in creating matte paintings to

manufacturing industries like using machines and cranes for sophisticated and repetitive camera or set movements. He explains, "This trend that visual effects borrowing technology from other fields is something that we continue to see to this day" (DNEG, 2017). *Figure 5* shows how the digital 3D models and environments in *Blade Runner 2049 (2017)* replaced miniatures used earlier to create cityscapes in *Blade Runner (1982).*

Bolter and Grusin (1999) point out that "remediation is both what is "unique to digital worlds" and what denies the possibility of that uniqueness" (p. 50). There is an ongoing debate that as long as digital cinema continues to rely on techniques developed during the analogue era, it cannot fully establish its own unique aesthetic (Niu, 2020). In a recent interview, Béla Tarr (in Partizán, 2023) discussed his expectation in the past that the emergence of digital technology and video cameras would lead to a new form of expression and language:

> In my opinion, video is not film. It is a new form of expression. As the saying goes, film is the seventh art. I thought that there would be a new eighth art, a technology-based visual form of expression. This is what didn't happen. Now 90% of people use electronic devices as trade-ins for film cameras. The form of expression hasn't changed (Partizán, 2023).

According to Rodowick (2007), digital cinema only slows down its own aesthetic development when it tries to imitate traditional cinematography. Despite the change of the medium, the narrative conventions of cinema have remained largely unchanged. He further argues that to fully understand the complexities of modern cinema as a new medium, we need new concepts that recognize the relationship between the three historically distinct strands of photography and film, electronic imaging and transmission, and computational processes (Ibid).

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achieve a more realistic combination of painted backgrounds and live-action shooting. This effect is particularly useful in creating shadows and ensuring proper lighting. Source: Spencer Meyer (2014, May 27).

Two Tiers of VFX

"There are two big threads of digital technology that have blossomed up.
The big effects ones and then the ones that every film can use a little bit of,
and you don't even know there's stuff there a lot of the time"
Jim Morris[8] (in Light and Magic, 2022b).

Hidden / Invisible effects

The practice of imitating traditional film language in VFX is commonly known as invisible effects; their goal is "to be mistaken for a live-action plate" (Giralt, 2017, p. 20). This refers to computer-enhanced scenes, a combination of digital and live-action plates, designed to deceive the audience into believing that the shots were filmed in reality (Manovich, 2002; Prince, 1996). Nicolas Chevalier, VFX supervisor of *The Revenant (2015),* explains that "the range of invisible effects included digital matte paintings and set extensions, fluid and particle FX simulations, character animation, lighting, CG trees, cleaning and compositing" (Frei, 2016). Almost 80 per cent of the film was digitally altered, meaning that most of the shots were completed using digital technology after being recorded rather than being captured directly by the camera (Lamm, 2018).

Forrest Gump (1994) was a pioneer in the realm of early photorealistic computer-generated imagery (CGI). "If it carries through an exciting moment that blends beautifully with the live action, that's what it's about" – says Ken Ralston, VFX supervisor of the movie, and he explains that digital cinema enabled a different type of filmmaking that made things possible on screen, and some movies are designed to use invisible visual effects (Light and

[8] President of Industrial Light and Magic 1994-2005, the quotation is from the tv miniseries Light and Magic season 1 episode 6

Magic, 2022b). *Forrest Gump (1994)* won the 67th Academy Awards for visual effects competing with movies that are more associated with CGI work, like *The Mask (1994)* and *True Lies (1994).* The movie received critical acclaim for its remarkable work in creating a convincing portrayal of Gary Sinise's character, as an amputee by digitally removing his legs with the help of blue stockings, which today may seem very easy to achieve but at the time it was kind of a sensation (Pooley, 2021). As *Figure 6* illustrates, to create an extra layer of realism to the scenes, the filmmakers were paying attention to such details as adding extra physical obstacles to the direction of the fictionally non-existing legs (such as a small table or the hospital bed itself) to convey that what we see could not have happened otherwise. “This is the kind of effect that could be never done with film; there are too many subtle blends, too many nuances in the shot that you can only do in digital” (Ken Ralston in Aubry, 2009).

Figure 6 *Invisible visual effects in Forrest Gump (1994). Source: Aubry Kim (Director, 2009) and Light and Magic (season 1, episode 6, 2022).*

David Fincher has mastered the use of hidden effects. He avoids practical blood in violent scenes and instead opts for CGI splatters, which allows for recording multiple takes without wasting time on cleaning the set, so he and the actors can concentrate more on the performance (kaptainkristian, 2017). Furthermore, in *The Girl with the Dragon Tattoo (2011),* he used VFX to maintain continuity of the little gap in Rooney Mara's bangs, adding it digitally to every shot. Furthermore, he applied face replacement in a motorcycle chase scene to capture the character's emotional state *(Figure 7);* however, it would be much easier to keep the stunt riding on the motorcycle in a helmet (kaptainkristian, 2017). Kirk Baxter, the editor of *Gone Girl (2014),* discussed his experience collaborating with David Fincher on the film's visual effects:

> He stabilizes almost every shot. Not really computer stabilizing, but by hand, pinning where you want it to move, stop, and reframe. He throws bluescreen everywhere you see, out of every window. So, there's an enhancement of everything, even if it's not just strictly an effect shot (Hullfish, 2017).

Figure 7 *Face replacement in The Girl with the Dragon Tattoo (2011). Source: (kaptainkristian, 2017)*

Fincher prioritizes storytelling over flashy CG effects, using VFX only to enhance his narratives. He aims to immerse the audience rather than impress or show off (Dry, 2017). *The Social Network* (2010) does not seem to be a particularly effect-heavy film, but it has around 1000 VFX shots, while *Godzilla* (2014) has only 960 (kaptainkristian, 2017).

Transparent Immediacy. The theory of transparent immediacy refers to a medium attempting to remove or hide its existence, a "style of visual representation whose goal is to make the viewer forget the presence of the medium (canvas, photographic film, cinema, and so on)" (Bolter & Grusin, 1999, pp. 272-273), and the desire to “get beyond the medium” (Ibid, p. 83). In visual effects, the primary objective is usually to make the digital modifications appear seamless and as realistic as possible as if the camera were capturing the scene naturally. In addition, VFX can serve the purpose of immediacy when removing stunt wires or other unwanted elements and restores a film's unbroken surface and its transparency (Ibid).

A key technique to achieve immediacy and to create a realistic representation of the natural world is understanding how to translate our 3-dimensional world into a 2-dimensional surface. (Ibid). Since the Renaissance, painters have tried to achieve immediacy through mathematical techniques for measuring space. Linear perspective, projective geometry, and Alberti’s Window[9] became a technique to create correct representations of the world on a flat surface. Automating the technique of linear perspective (camera obscura, photography, film) was seen as the ultimate and perfect realisation of perspective. A similar technique is used in digital 3D rendering; thus, like perspective painting, digital graphics rely on mathematics to describe nature (Ibid).

[9] "On the surface on which I am going to paint, I draw a rectangle of whatever size I want, which I regard as an open window through which the subject to be painted is seen” (Alberti 1972, as cited in Bolter & Grusin, 1999, p. 25).

Obvious / Visible effects

Visible visual effects are intentionally designed to be noticed, like explosions, natural disasters, supernatural powers, environments, and creatures that obviously could not exist in the real world or would be too difficult, impossible, or dangerous to capture on camera. These effects are typically created by using computer-generated imagery (CGI) to add spectacle to a movie. For example, in *The Revenant (2015),* when Leonardo DiCaprio is fighting with a bear, or in the case of *Life of Pi (2012)* with an actor sitting in the middle of the ocean in a boat with a Bengal tiger, it would have been extremely dangerous and impossible to shoot practically. Likewise, the world of the Marvel Cinematic Universe would be hard to mistake for anything based on real life (Lamm, 2018).

Giralt (2017) suggests that the latest digital technology in the entertainment industry led to a certain scepticism, and CGI or VFX became an interchangeable term with ‘fake’. John Nelson, VFX supervisor of *Blade Runner 2049 (2017),* shares his philosophy about visual effects, which is a triad of “*making it look real, making it look good, and to echo the story* […] And if you violate any of those, it sort of falls apart” (BAFTA Guru, 2018). He explains that too many times, CGI is just done to add spectacle and wow effect. “I only use all of that engineering or math skills in the service of trying to make an elegant solution to a complex problem” (BAFTA Guru, 2018).

Figure 8 *Example of a visible, hypermediated visual effect. Source: Blade Runner 2049 (2017)*

Hypermediacy. Hypermediacy is the opposite approach to immediacy when a medium draws attention to itself and emphasises its own medium-specific features (Bolter & Grusin, 1999). In cinema, hypermediacy can be achieved through montage, split screens, and other optical and visual effects, extreme camera angles, exaggerated lighting, surrealistic imagery and fantastical creatures and characters that emphasise artificiality that draw attention to the medium and disrupt the illusion of transparency. This approach is often used in films where the special effects are meant to be a central aspect of the film's style and aesthetic, such as in sci-fi, action and fantasy films (Ibid).

For example, the film *Vertigo (1958)* portrays the protagonist's vertigo attack using the track-out/zoom-in camera technique to achieve a particular visual effect that interrupts the otherwise transparent film style, and the audience suddenly becomes aware of the medium (Bolter & Grusin, 1999).

Or in *Terminator 2 (1991),* viewers are intended to notice the computer graphics used for the evil 'mimetic poly-alloy' character to change its shape *(Figure 9).* The film walks a fine line where the graphics must be remarkable and believable. The audience is not expected to believe that the camera recorded the supernatural scenes in reality; instead, they are asked to believe that if such a robot existed, its visual transformations would be represented as they see it (Bolter & Grusin, 1999).

Figure 9 *Another example of a visible, hypermediated visual effect shot. Source: Terminator 2 (1991)*

Integration of Spectacle in the Narrative. While the desire for immediacy has dominated Western representation since the Renaissance, hypermediacy has often been relegated to a secondary position. In contemporary times, hypermediacy can be seen as the opposite of immediacy and as an alter ego that has never been completely suppressed (Bolter & Grusin, 1999). Immediacy and hypermediacy can be present in movies simultaneously, while the audience is aware of the gap between 'what it knew and what it saw' and 'the looking at and the looking through' while becoming conscious of the desire for immediacy (Ibid). In *Jurassic Park (1993),* the audience is aware that the dinosaurs are not real; however, they wonder how the interaction with the live actors can be so realistic. "We go to such films in large part to experience the oscillations between immediacy and hypermediacy produced by the special effects" (Bolter & Grusin, 1999, p. 157). Bolter and Grusin (1999) further suggest that hypermediacy or the awareness of the medium is part of the amazement. If the audience forgets the medium, it will undermine the amazement and wonder. If the viewer does not realise that what they see is constructed or mediated, they will not experience the same sense of surprise or delight when they realise that they have been ‘fooled’. Therefore, the amazement requires hypermediacy, which can be present in a transparent film style, “so the double logic of remediation is complete” (Ibid, p. 158).

Gunning (1990) describes the perception of a spectacle during the years of the ‘cinema of attractions[10]’ in a way that the spectacle dominates over the narrative, the shock and surprise work against the story, it halts the narrative and the diegetic universe. “The spectator does not get lost in a fictional world and its drama, but remains *aware of the act of looking*, the excitement of curiosity and its fulfilment” (Ibid, p. 57). The cinema of attractions or a spectacle does not care about the psychological motivation or the character's personality; the “energy moves outward to an acknowledged spectator rather than inward towards the character-based situations essential to classical narrative” (Ibid, p. 57).

Laura Mulvey (1975) identified the classic Hollywood female character as the object of desire, whose presence is often represented as a spectacle that "freezes the flow of action" (p. 11), which is similar to Gunning’s description. She observed that these moments are integrated into the narrative through the look of the protagonist (male gaze) without breaking the diegesis or narrative verisimilitude. Mulvey mentions the three distinct looks in cinema: the look of the camera recording the pro-filmic event, the look of the spectator watching the movie, and the look of the characters within the diegesis (Ibid). As the spectator identifies with the protagonist, when the look of the protagonist and the look of the spectator meet on the spectacular shot, it ensures the spectator's absorption in the narrative event (Ibid).

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[10] Gunning referred to the “cinema of attractions” as the first decade of cinema until around 1904-1906, which purpose was providing the marvel of moving image, before the narrative conventions of cinema were solidified.

Mulvey's concept of the male gaze can be translated to the previously mentioned *(Figure 3)* and iconic scene in *Jurassic Park (1993),* where a giant CGI dinosaur was first revealed on screen (Film & Media Studies, 2021b). The delivery of the spectacle is preceded by a prolonged build-up created through the characters' looks. The audience finally sees the spectacular CGI dinosaur only after the camera lingers on Sam Neill's astonished face and then shifts to Laura Dern's stunned expression, emphasising *the act of pure looking*. This editing decision of the order of the shots heightens the suspense and anticipation before the reveal (Film & Media Studies, 2021b).

Figure 10 *The 'Spielberg-look' in Jurassic Park (1993)*

The 'Spielberg-face' is a recurring element and one of Spielberg's signatures. The camera stays long on the characters' faces, that conveys that something monumental is happening and tells the audience to feel and react accordingly before the reveal (Lee, 2017). The character surrenders in the *active looking*, mirroring the spectator's action (the act of *active looking* in the cinema). Dwelling on expressive looks on faces reacting to an event off-screen is a common device and has long been established in Hollywood cinema, often accompanied by zooming or dollying in on the close-up of the faces (Ibid).

Editorial Workflow with VFX

Editing and special/visual effects have a close relationship and intertwined histories; however, they have developed into separate disciplines, and they are meant to help each other. Visual effects help to create certain types of transitions, and editing helps to maintain the credibility of special effects and to hide its artifice. For instance, with the increasingly rapid cutting pattern, editing helps to achieve the realism of fantastical creatures (Keil & Whissel, 2016). In *Jaws (1975),* finding the perfect length for the shots of the animatronic shark helped to make it look more realistic and scary. As Spielberg recalls:

> The sad fact was that the shark would only look real in 36 frames and not 38 frames. And that 2-frame difference was the difference between something really scary and something that looked like a great white floating turd (Ibid, p. 113).

From Storyboard to Turnover

While editing traditionally started after or simultaneously with principal photography, with the computerization of the visual effects, it is part of pre-production as well (Keil & Whissel, 2016). The process of developing visual effect sequences typically begins with storyboarding. Editors usually use the storyboard cells as placeholders later in the edited sequences and sometimes edit, animate, and move them to imitate the pace and duration of the action. Visual effect companies are making a so-called previs[11] for shots or complete sequences, and editors often edit them as part of the "very origin of the editing process" (Ibid, pp. 19-21). The editor can start building the sequences and work with the director to decide what is necessary or omitted before filming begins (Assemble, 2022).

[11] Previs is a basic animation, a not too detailed visualisation of a visual effect shot.

During principal photography, live-action plates are recorded, which can be combined with digital assets in post-production. During the editing process, decisions are made about the selection of plates and elements, their duration, pace, and position relative to each other.

Postvis is a combination of the live-action plates and the different VFX elements created by the vendor[12] or a postvis team (Cheung, 2023; Garcia, 2022). By using postvis, studios can better understand the scope and complexity of a project, allowing them to provide more accurate bids and potentially save time and money. Also, the editor can work with the postvis while final visual effects are being made (Failes, 2018). The editorial team may take on postvis work by creating rough visual effects, so-called 'temp comps' typically handled by the assistant or VFX editor (Hullfish, 2017). Temp comps provide a visual reference for the editor and are essential for internal screenings until the final shots arrive (Garcia, 2022). While these comps do not need to be highly sophisticated, they must be good enough to effectively tell the story without disrupting the viewer's immersion in the film.

As part of the editorial team, the VFX editor plays a crucial role in overseeing the visual effects workflow, ensuring that the editor works with the most up-to-date versions of previs and postvis for each shot by placing and timing them properly into the cut (Hullfish, 2017). It is essential to keep track of every shot to monitor its status and make changes as needed or adding or removing a shot. This task requires excellent organisational skills and meticulous paperwork by building and working with databases (Garcia, 2022).

Once the editing process is finalised, the shots that include visual effects must be handed over to the vendor. This involves packaging all the necessary elements, such as plates, assets, foreground and background and other materials for the shot, and a copy of the full sequence and the corresponding count sheet[13]. This process is crucial in ensuring the vendor has all the necessary information and materials to complete the visual effects work accurately and efficiently (Garcia, 2022). The entire process must be monitored until

[12] VFX company
[13] The count sheet is a detailed document that provides information about each VFX shot, including the number of frames, the type of elements used, and any specific notes or instructions.

the final stage of colour grading (DI), and a correct colour management process must be followed (Hullfish, 2017).

The editor needs to find a balance between starting VFX work on shots at full resolution (turning them over to the vendor) and waiting until the cut is finalized. This is important because any work done on a shot later dropped or changed in the final cut will still need to be paid for, in addition to the cost of any new shots added to replace the dropped ones. This balancing act can be challenging, as VFX shots are often needed for test screenings or finished in time for trailers, and the vendor needs enough time to complete the work (Lark, 2022).

Composite as an Editorial Tool

It is also worth noting that some editing tools are difficult to replicate in the DI[14] process, so some cuts, such as split screens, re-speeds, re-sizes, and re-framing, may become VFX shots included on the VFX shots list (Lark, 2022). VFX editor Eddy Garcia[15] (2022) explains that most people think of visual effects as giant CGI creatures and green screens, but it is much more than that; they often go unnoticed to improve or solve different narrative problems. During his masterclass, he gave examples of how easy fixes with VFX can help the editors work.

Simple shots can be created by combining elements from different sources or takes, and the timing can be adjusted to achieve the desired effect and improve the edit's overall flow. The most common technique is to use performances from different takes in different parts of the image, which happens much more than an average viewer might think (Garcia, 2022). He demonstrates it with a scene from *The Unforgivable (2021),* where

[14] Digital intermediate (DI) is a motion picture finishing process which classically involves digitizing a motion picture and manipulating the colour and other image characteristics. Effects that can be done in DI is typically: simple cut, dissolves, crossfade, repositioning and reframing. (DI - Effects, n.d.)
[15] Eduardo Garcia, VFX editor, worked on movies such as Ad Astra (2019), Dune (2021), Godzilla vs Kong (2021) The Marvels (2023), and Dune Part Two (2023) among many others.

the shot's left and right sides are combined from different takes, and the image is divided in the middle – a so-called split screen *(Figure 11).* It gives greater control over the narrative, and the editor can assess performances separately. This is an especially useful technique applied for over-the-shoulder shots to maintain continuity or adjust the pacing of the scene.

Figure 11 *Split screen as an editorial tool. Eddy Garcia (2022) demonstrates the combination of different takes in one image through an example from The Unforgivable (2021).*

Furthermore, with editing, it was always possible to shape an actor’s performance by selecting and cutting. Now “digital technologies allow editors to edit within a shot to make minute changes to a performance” (Keil & Whissel, 2016, pp. 19-21). For example, adjusting the speed or freezing certain elements or characters helps to eliminate any unnecessary movements or to retime them, as demonstrated in *Figure 12*. This technique can be extremely useful in extending the duration of a shot or adjusting the right pacing (Garcia, 2022).

Figure 12 Freezing or re-speeding elements in the frame to achieve perfect timing. Source: The Unforgivable (2021)

Furthermore, a simple and realistic shot can be combined with multiple layers and elements, creating a naturalistic and well-paced rhythm that looks like it could have been captured in front of the camera (Garcia, 2022). Or adding elements, like cars passing through in the foreground, can further enhance the shot, depending on the need of the story (Garcia, 2022).

Figure 13 A shot can consist of multiple elements. Source: The Unforgivable (2021)

In cases where story problems arise, adding or creating new shots is another effective solution. For example, Garcia refers to the film *Papillon (2017),* where a shot was required to show a character seeing a couple kissing. Instead of reshooting with the challenge of replicating the same environment, the problem was resolved by combining a clean plate from the original scene with a separate shot of the character (Garcia, 2022).

__Figure 14__ Solutions for narrative problems with VFX. Source: left: Papillon (2017), right: Ad Astra (2019)

Similarly, in *Ad Astra (2019),* a character was removed from the film entirely. However, the filmmakers wanted to keep a scene that had already been shot with her. To solve this issue, they replaced the original character with another one added into the scene (Garcia, 2022).

VFX can also be a helping tool to maintain continuity between the shots. Garcia (2022) gives an example of adding a reflection to a car window to match the action with the previous shot when another car was passing by. All these adjustments described above were not necessarily planned; they came up during editing as a different way to solve story problems, enhance the shots and the rhythm or benefit clarity.

III. Practical Approach

This chapter focuses on the narrative qualitative research method, specifically on the data collected through an interview conducted in December 2022, with Joe Walker, ACE, a renowned film editor known for his work on critically acclaimed films such as *12 Years a Slave (2013, Oscar nominee), Arrival (2016), Blade Runner 2049 (2017), and Dune (2021, Oscar winner)*, among many others. During the interview, Mr Walker discussed his experience working with visual effects in contemporary Hollywood films. The insights gained from this interview provide a valuable opportunity to understand better the editing approach to visual effects and the collaboration between editors and VFX artists in contemporary cinema. Additionally, this research aims to identify the new challenges film editors face and the opportunities digital imaging can offer to improve editing and storytelling.

Editorial Workflow with VFX

Joe Walker (personal communication, December 3, 2022) discusses that an editor's involvement is certainly longer for VFX-heavy films. He starts earlier than usual to edit storyboards and previs, collaborating with the director to ensure the shots are appropriately budgeted and planned. He explains that it is a good way to see if something is missing or unnecessary, and if there is a chance to remove shots, it is better to know early.

Additionally, after the director's cut, there is still a lot of VFX work to be done, whereas typically, as post-production goes on and the film moves into the grading stage, the cinematographer becomes more involved, and editors less – but with VFX that involvement is longer. Walker points out that in the early days of VFX, editors received standalone shots from the special effects department and had little involvement in combining them, whereas now it is a much more collaborative process, and an editor is hugely involved in these decisions.

He further explains how using storyboards and previs affected his work on films like *Blade Runner 2049 (2017)* and *Dune (2021).* Storyboards are like a shopping list for shots but do not always indicate dialogue or shot repetition. He often edits the storyboard cells, indicating dialogue and sound design:

> I'll pan them and move and crossfade them and try to kind of give a sense of the timing […] trying to figure out the story before it was shot. I spent a lot of time on something that nobody's ever going to see (Walker, personal communication, December 3, 2022).

The previs is more to help set design, VFX, and stunts, and it is a good guide for them to budget and prepare, but on set, they shoot according to the storyboards, and the director and the cinematographer will adapt to the real-world situations they are facing on set.

> Denis's [Villeneuve] mantra is: there's the script, the previs and the storyboard. And then there's the footage, and the footage is king. And then the storyboard and then the previs and then the script (Walker, personal communication, December 3, 2022).

Editing Highly Pre-Conceptualized Scenes

Walker explained that despite the well-planned pre-conceptualisation, the editor’s role is still crucial because they bring an element of time, and they have to make fundamental decisions about where things are on the screen. He gave an example from a recent project where they had to create the illusion of a genuine relationship between two characters who were not on screen together. The scene features a huge fictional sandworm and two characters, Paul and Stilgar. They had a shot of the desert with one character in the foreground and the other riding on the sandworm, a CGI creature that was not actually there. The editor had to combine two separate shots to create the illusion of the worm moving through the desert with Stilgar riding on top. Also, he had to decide how fast the worm would believably move (by moving the live-action plate of the character relative to the framing of the shot) and which character to wave first to the other, determining their relation in the context of the narrative. In contrast with a conventional shot where the relationship between the characters is already established and captured on camera, “You wouldn't have a choice. You'd have whatever they shot on the day” (Walker, personal communication, December 3, 2022).

Cutting with a Missing Element

Working with visual effects, editors often need to wait until the final shots are delivered, and they lack visual cues for pacing and determining the next shot, which poses a challenge in the process (Hullfish, 2017; McLean, 2007). In these cases, the editor needs to use temporary placeholders, like a title card with a description, storyboard cells, or previs elements that demonstrate motion and timing. Music, sound effects or voice-overs may also be used (Paquette, 2022).

Walker reflects on his experience working on the film *Arrival (2016)* and the scene when Amy Adams’ character puts her hand on the glass inside the spaceship, and a Heptapod[16] puts its hand on the other side of the glass *(Figure 15),* and obviously, there were no aliens recorded on set, which made it difficult to determine how long to hold certain shots. “I haven't got anything to put there, but I can't just hold a white screen. Because everybody's going to doubt whether we want to be that long on the shot” (Walker, personal communication, December 3, 2022).

[16] the alien species featured in the film

To mark out timing, the sound designer developed a believable vocalisation of the Heptapods, which had articulation and syntax, rather than just using whale noises or other generic animal sounds. This allowed them to structure the sequences without the CGI creatures and understand how long to hold each shot. “That's my job, interfacing visuals with time. And my strongest ally I've gotten that is sound” (Walker, personal communication, December 3, 2022).

Figure 15 *A CGI creature that was not present in the dailies. A shot from Arrival (2016) and a behind-the-scenes image for the corresponding scene. Source: Paramount Movies (2017)*

Additionally, the editorial team used parts from the storyboard to further develop the shots and provide temp comps as timing guides for the VFX team. With the help of the VFX editor, they could produce something good enough for internal screenings, even in a large cinema, that is convincing enough so that feedback can be received, and changes can be made before it is too late.

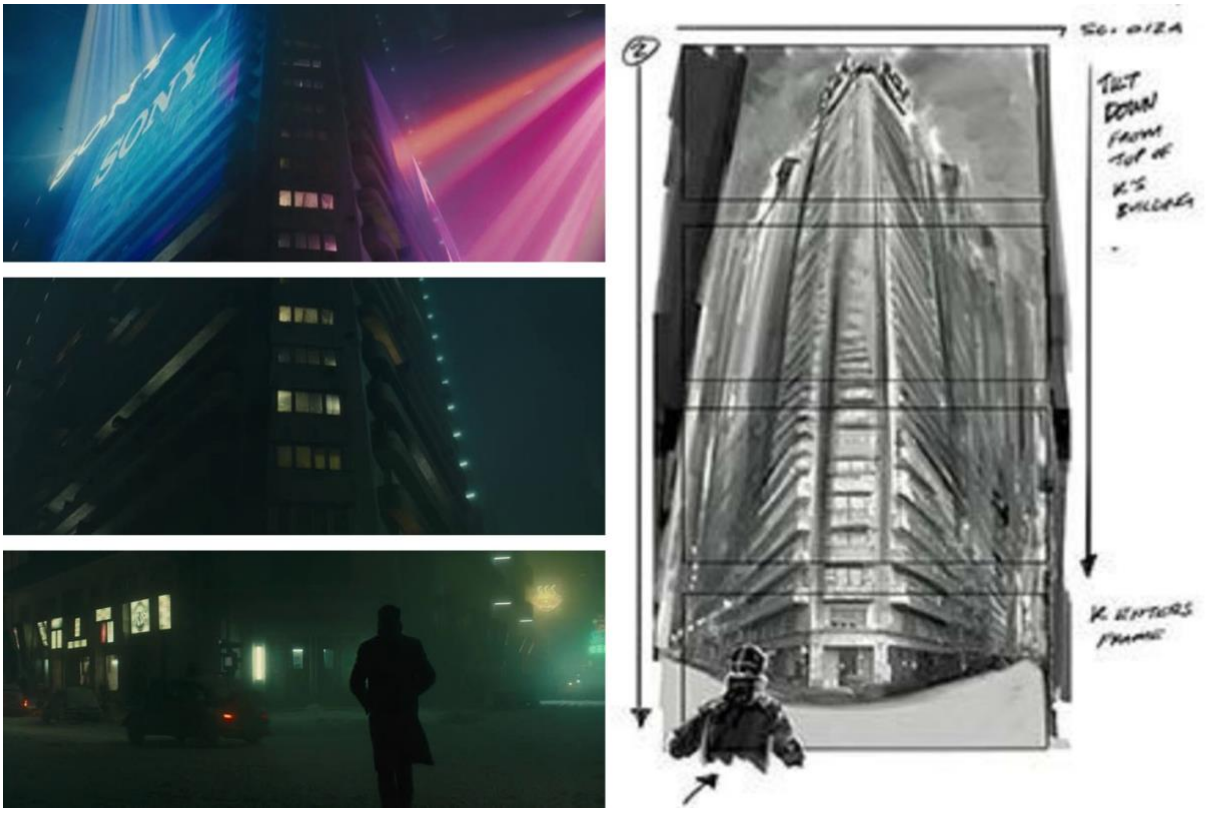

Figure 16 *The apartment tilt-up shot from Blade Runner 2049 (2017). Images from the movie and the corresponding storyboard cells source: Hudecki, (2021).*

Walker brings another example for creating the pace of certain shots that are CG built, and some parts are not present in the dailies. In *Blade Runner 2049 (2017), when Officer* K walks into his apartment building:

> And when the camera tilts up, there's no apartment block so I could work out how long we pan out for. How tall is the building? You wouldn't know without concept art because the plane just pans up, and it doesn't have an end. It just goes up into nothing (Walker, personal communication, December 3, 2022).

As a solution, the editorial team had to integrate the concept art and storyboards elements into the live-action plates to determine the final length of the shot.

In *Dune (2021),* the scene with the hunter-seeker[17] was also challenging regarding working with an essential component missing. In the early stage, Walker had to work with the empty plates, which contained some camera movements that gave him a cue for the motion of the hunter seeker.

> I'm aware of the storyboard, the previs, and the action finds rhythm and time in my hands. So, you're trying to find a way to determine how long to hold for this thing, especially because it's really creepy when it's slow. We wanted something very sinister and to take our time to show Paul seeing it and sweating. I ended up using some texts with the title tool, and I had the word "hunter-seeker" moving, and if I may say [laughs], I outdid myself by using perspective, which I'd never done with the title tool before. Anyway, every shot had the word "hunter-seeker", going past very slowly in the foreground" (Walker in Hullfish, 2021).

Figure 17 *Paul faces the hunter-seeker in Dune (2021)*

[17] A tiny floating mosquito-like deadly, poisonous machine used by assassins, remotely controlled by an operator. In the mentioned scene, Paul Atreides, the protagonist is attacked by the hunter-seeker.

Two Tiers of VFX

Composite as an Editorial Tool

Walker discusses the distinction between complex visual effects and simple editing techniques, such as speed adjustments, split screens or reframing. He suggests that these simple fixes should not be considered visual effects because they are relatively easy and inexpensive, while he acknowledges that these techniques can significantly improve shots. These are a key part of creating the pace and the rhythm and making the best out of the dailies. He further explains that if a shot is already a visual effect shot, it is even easier to manipulate and optimize the plates and “knitting together two or three little bits.” The amount of these little digital fixes has exponentially risen over the years, “It is not very Dogme95, I know” (Walker, 2016).

> But somebody would feel very cheated if they were told that, Oh, yeah, *12 Years a Slave* is amazing for VFX. […] I think there are two tiers of the effects. There's the really elaborate creative design work, and then there's the kind of functional editing VFX, which I think we should stop thinking of as VFX, personally. […] If I've got a fantastic performance on the left-hand side of the screen and a fantastic performance on the right-hand screen, there's some nice little way where I think the two shots can be bolted together. Of course, I'm going to do that. I can save a shot; I can use the take I want to use. And I always think that's part of an extension of the choices I make in editing. I wish they belonged to the editorial department (Walker, personal communication, December 3, 2022).

Walker suggests that digital artists should collaborate closely with the editorial team to create the final shots rather than outsourcing them to the VFX department, as it can be bureaucratic, with too many channels to filter through. By adopting this approach, the workload of the VFX department could be reduced, and post-production costs could be minimised.

Integration of Spectacle in the Narrative

> "It doesn't matter how great the world-building is; if you don't care for the central characters, then we might as well go home."
>
> (Walker in Hullfish, 2021)

Walker discussed using visual effects in *Arrival (2016)* to explore how to deliver a spectacle while supporting the narrative. He referenced the highly detailed CGI spaceship that was rendered in every centimetre of its structure but was intentionally kept out of focus in the background to create a sense of grief, tension, and realism. For a long time, the spaceship was shown on small, out-of-focus television screens in the background instead of showing off with a detailed CGI structure. "It must be very irritating to the VFX artists to kind of go to the trouble of making this fantastically detailed spaceship and then choosing to put it in the depth of focus" (Walker, personal communication, December 3, 2022). Villeneuve's cinema aims to keep the audience's imagination active and relies on glimpses and suggestions to land in the audience's imagination.

> This is a very different approach from 'kids' science fiction' where every transformer has a million little contraptions that you can see in perfect sharpness at all times, which is not about imagination. It's about kind of boys and toys (Walker, personal communication, December 3, 2022).

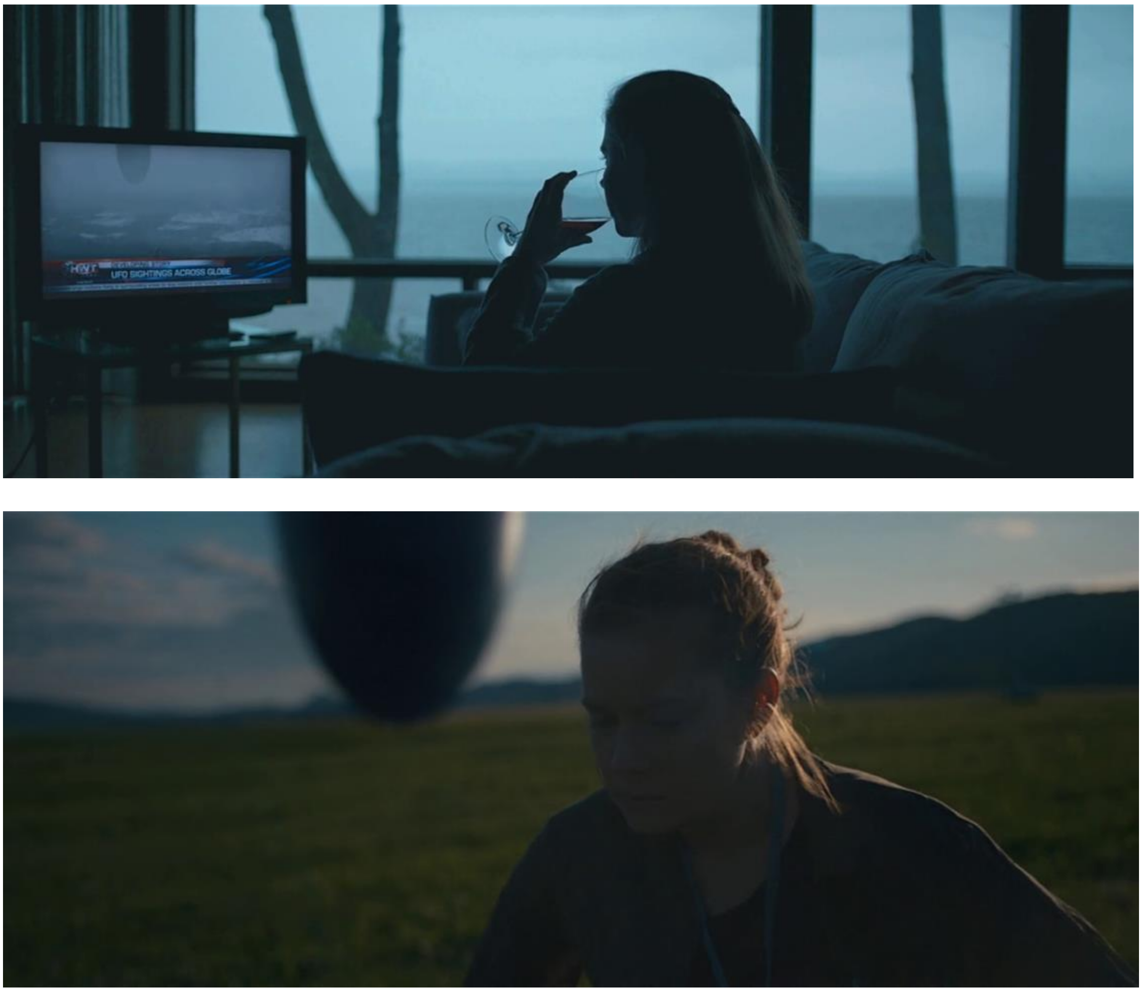

Figure 18 Realistic presentation of a CGI element in Arrival (2016)

At the same time, Walker explains that delivering a spectacle in movies has its own importance and purpose in the narrative. His description closely aligns with Bolter and Grusin’s (1999) observations about the oscillation of immediacy and hypermediacy in film. He praised Villeneuve's ability to understand the importance of suspense and not to show everything but to build up the critical moments in the film:

> That's actually about not having any VFX at all. It's about setting it up right, and when really delivering the spectacle, you get to see it. That's the joy. When we go out into the desert [in *Dune (2021)*] for the first time, everything is physical effects, like the vibrating sand. It's a bit like the vibrating cup in Jurassic Park. You feel the thing coming for a long time before you actually see it. And then when you do see, yes,

you better deliver a spectacle because that's what people have spent $16 on going to see. And a babysitter. That's why they're there in the cinema” (Walker, personal communication, December 3, 2022).

Synthetic Realism

“Cinema is an illusion that 24 dead images run past the eye
and give you an appearance of motion, which isn’t happening.
I mean, it's all fake. Cinema is fake”
(Walker, personal communication, December 3, 2022).

With his storytelling instinct, Villeneuve, in his movies, effectively used VFX to create fantastical and impossible elements in a plausible universe. During a Q&A session with Eric Roth (Dune’s screenwriter), Mark Mangini (sound editor), and Theo Green (sound designer), they mentioned that their approach was to create a kind of documentary about the fantasy world of Dune. This resulted in a new aesthetic in science fiction known as "fake documentary realism" (FDR), which Green coined as a term. Unlike most sci-fi films that rely on electronic synthesizer sounds, Dune's FDR aims to create a realistic visual and auditory experience to immerse viewers in the story's world. The goal is to mimic realism that the audience believes the world of Dune is real. This aesthetic is evident not only in the film's sound design but also in its CGI work, cinematography, performances, and script structure (Eiselstein, 2022).

Walker explains the concept of FDR as making the sounds feel as if they were recorded with a microphone placed near the object, a spaceship or the Heptapods. This approach embraces imperfection and adds realism to a shot, creating an aesthetic that mimics documentary. This is similar to the visual experience of realistic visual effects as well.

> On the set of Dune, they just cover everything with thousands of tons of dirt. And it's sort of embracing the kind of imperfection of shooting into the sun and having a lot of aerial halation and all these things that add realism to a shot. That's the aesthetic (Walker, personal communication, December 3, 2022).

Compositing

Walker compares image compositing to musical composition, which involves finding the pace and rhythm of the various components. He always tries to create a pleasing pattern for things that may or may not be like real life. Working on a VFX film gives more room for creating an ideal outcome; it gives the editor and director more control.

> I have a debate in my mind about whether I'm overly controlling things. And sometimes, I wish I would allow accidents, but I think I go the other way. I take things and try and arrange them to make them look like an accident, but they're not. It's strange. […] If you've got a shot which has three or four components, then you get to play with the timing of each individual. You kind of create the illusion of a rhythmic piece that might have been captured in reality (Walker, personal communication, December 3, 2022).

He gives an example of a complex scene in *Blade Runner 2049 (2017)*, which required careful coordination of separate elements to create a cohesive sequence. He mentions that the ‘hologram show scene[18]’ was challenging to make all the different layers and elements work to convey the narrative purpose. “If you imagine it's harder to do that than it is to just start from fresh with three actors in an empty room, we had to fit our ingredients in visually,

---

[18] It is the scene where Officer K, the main character hides in a ballroom and tries to evade Deckard, who has the upper hand because he's familiar with the holograms that will appear. The holograms are intentionally dysfunctional, with glitches and distortions, making it a "broke" hologram show.

but also rhythmically and against music” (Walker, personal communication, December 3, 2022). Eventually, the scene was rethought entirely to make it work better within the context of the film.

The director originally planned a scene with multiple layers of cowboy music, Folies Bergère, Liberace, Marilyn Monroe, and other iconic 20th-century references. However, the music box had broken, resulting in all the sounds playing together in a jumbled mess, a horrible noise. After a long and meticulous process of assembling this scene, the producers expressed satisfaction with the overall result of the movie, except for this scene - which they felt did not fit into the film and had lost its intended tension and suspense, and ended in a show-off kind of magic show, detracting from the storytelling. The studio asked them to cut out the scene, instead, they changed their approach and focused on adjusting the music and sound effects:

> Let’s dump the music and just have a little bit of Elvis popping in from time to time, a tiny bit of Folies bergère, but just never a whole piece of music. And so, it becomes more violent the way it turns on and off. We got rid of things that were going to be a passing tiger, that was going to be a lot more acrobatics. So, we kept sort of the spirit of it, but it just made it much tenser (Walker, personal communication, December 3, 2022).

The sound editor, Theo Green, played a significant role in enhancing the scene by providing unique sounds like creaking and heating from the lighting rigs in a theatre. Ultimately, the changes they made saved the scene from being removed entirely.

Figure 19 The 'Hologram show scene' in Blade Runner 2049 (2017)

Walker emphasises the importance of treating the finished VFX shots like any other, disregarding the cost and effort that went into creating them to ensure it fits seamlessly into the final product and serves the story. In the same movie, the filmmakers had a scene involving a character, Rachel, who had to convincingly look like the actress Sean Young, who originally played this role in the 1982 movie *Blade Runner*. However, 35 years later, the actress no longer looked the same, so the filmmakers used a stunt imitating the movements of Sean Young, and later they replaced her face. Today's technology that we take for granted, with deep fakes and easy human face replacement, did not exist at that time. So, they used CGI animation. The VFX artists spent almost a year on the process, scanning hundreds of images and creating a 3D model. But animation and real humans behave and look differently on screen, and the originally planned sequence did not look convincing enough. So, the director suggested, “Why don't you just treat this as if it was dailies? And cut it as you would do. And you don't have to stick to the original timing” (Walker, personal communication, December 3, 2022), despite the high cost of each frame. Ultimately, this decision helped the film and made the sequence better, more realistic, and naturalistic to the eye***.*** This serves as another example besides the earlier mentioned *Jaws (1975)* when editing and finding the proper duration of the shots supports the realism of the visual effects.

> It feels devilish because […] you decide to cut a shot out, and you're cutting a sizeable chunk of money out of the film. We did make changes, and I think it helped the film, helps that little sequence that we, having spent all that time and money, then ignored and just treated as if it was dailies day one (Walker, personal communication, December 3, 2022).

Figure 20 *Face replacement on Sean Young. Source: Blade Runner 2049 (2017)*

Another significant decision to make, in addition to shortening or removing existing footage, is to request new shots. For instance, during the *Dune (2021)*, the filmmakers found that without seeing a carry-all[19] functioning, it is not understandable when it is broken. As a solution, two computer-generated shots were added later to demonstrate the carry-all in operation. Although these shots were not part of the original plan, they provided crucial context to improve the storytelling.

Editing realist and formalist-style movies

About the difference in working on films with a more realistic style, Walker mentions Steve McQueen's films[20] as opposed to a VFX-heavy movie like *Dune (2021).* In a technical sense, McQueen's style focuses on using fewer shots and attempting to capture everything in one take or with minimal coverage, and the editing decisions focus on selecting the best takes. On the aesthetic side, Walker expressed his fascination with both the subtle, non-manipulative filmmaking approach, using long, uninterrupted shots, and the heavily managed viewpoint of more traditional Hollywood films. Additionally, he reflected on his work on the movie *Hunger (2008)* which included a 17-minute long take, and he compared the experience to looking at a Rothko painting, surrendering to it while searching for meaning, narrative and the “hand-holding” that a director normally gives to the spectator.

> But when you do reattach on a shot as you have in Hunger, there's a kind of real sense of your involvement, which is probably higher than it would normally be in just sitting, you know, eating popcorn, watching the latest show on HBO (Walker, personal communication, December 3, 2022).

[19] In Dune, „carry-all” is a vehicle that can lift and transport the huge spice harvesters when a sandworm is approaching and endangering it

[20] Joe Walker worked on Steve McQueen’s movies like *12 years a Slave (2013), Shame (2011), Hunger (2008) and Widows (2018)*

Overall, Walker was fascinated by the experience of working on a film with such a long, uninterrupted shot and the challenges and rewards it presented. He also found it ironic that he received a lot of attention for “not cutting.” “The first cut happens after 17 minutes, better be in the right place. It's a big decision to kind of not cut” (Walker, personal communication, December 3, 2022).

On the other hand, he praises the various technical aspects of filmmaking that contribute to this art form, a more “manipulative kind” of filmmaking, highlighting the importance of camera placement, shot length, and cutting rhythm in creating a cohesive and engaging cinematic experience. “They become such great rollercoaster ride and such a fantastic display of cinematic art, in all aspects” (Walker, personal communication, December 3, 2022). He further explains that editing a film with multiple layered foreground and background plates with multiple takes for each is a complex manoeuvre. Changing one shot in a VFX sequence is no longer a simple matter and requires rebuilding multiple layers.

> Sometimes on Dune, I'm spending 3 or 4 hours putting one shot into a sequence. And each shot had to have a lot of decisions around that multiplicity of decisions based upon the numerous elements of it. In simple terms, you're not just changing V1 and A1, you're changing V1-2-3-4-5 and the multiple temps that are above it and all the audio. I mean, it's a surgical job to change (Walker, personal communication, December 3, 2022).

Walker emphasised that both approaches are valid and important to the art of cinema, and most editors likely enjoy both disciplines. He also notes that it is essential to remain adaptable as a filmmaker because the subjects of films are different, and adaptability is a key trait for success in the constantly evolving field of filmmaking. As an editor, it is vital to "duck and dive" and adjust one's approach based on the project's specific demands: “I think that my dogma of saying I'm going to only hold shots for 17 minutes isn't going to last very long in a Mad Max film” (Walker, personal communication, December 3, 2022).

Collaboration with the VFX Department

Walker explains that while he tries to keep the hands off to a certain extent out of respect, there are times when there is an overlap between editing and VFX. This overlap can occur in the selection of shots, trying to maintain continuity between shots or sequences or ensuring that the pacing and speed of movement are consistent and effective. He further explains, similarly to Keil and Whissel's (2016) earlier mentioned observation, that the editorial and the VFX department are meant to support each other's work:

> You have to kind of bend for the other department to help each other out. And they similarly will hopefully bend and help us when there's something that needs to be fixed. We have to kind of, you know, you scratch my back, and I'll scratch yours (Walker, personal communication, December 3, 2022).

He acknowledges that editing and VFX are separate disciplines requiring different skills and expertise, but they also recognise the importance of working together to create a *cohesive final product. Collaboration is key, as each department brings its own unique perspective and expertise to the table.* "The VFX editor is my go-between the two worlds. I'm not technically skilled enough to know all the details of how they create this magic in the VFX department. But the VFX editor is there to help me" (Walker, personal communication, December 3, 2022).

However, the time pressure can make the collaboration challenging because VFX often requires a quick turnaround time. *"They always want it yesterday." He refers to the* earlier scene in *Blade Runner 2049 (2017)* with the face replacement. Walker had only one day after the shooting to decide and select the 'hero take' for the scene. He had to choose from 23 takes of Rachel walking forward and select the best one, as changing it later would be costly and time-consuming because the CGI-animated face would not fit all the takes.

> Denis [Villeneuve] always says: I can't hunt and cook at the same time. And by that, he means he's in gathering mode when he's filming. And similarly, I'm cutting in a different

> mode when I’m assembling and fine cutting. That's one of the hardest things about working with VFX, thinking of the fine cut while you're doing an assembly (Walker, personal communication, December 3, 2022).

He emphasises that collaboration and communication between the departments are crucial to getting the needed result when it comes to these kinds of difficult situations. However, picking the right shot early in the editing process was challenging. VFX needed almost a year to complete this shot, and they had to start working on it immediately.

IV. Conclusion

Computerization and the shift to digital have transformed filmmaking in ways that were unimaginable just a few decades ago. Due to the implementation of digital visual effects, filmmakers gained even greater control over the image during postproduction. The democratization of technology has made it possible for smaller and independent productions to incorporate visual effects that were once reserved for major Hollywood productions. Therefore, filmmakers need to adapt to more complex workflows and different approaches and practices.

Digital visual effects can be used for a myriad of reasons. They can go unnoticed and aim to achieve realism or to add fantastic elements and spectacle to a movie. The category of visible effects is typically used in genres like action, fantasy, and sci-fi. On the other hand, invisible effects are meant to stay hidden and to be mistaken with live-action footage, allowing filmmakers to realize scenes that are impossible, impractical, or dangerous to shoot live. I applied the theory of remediation from Bolter and Grusin (1999) along with immediacy and hypermediacy to the two main types of VFX. While transparent immediacy aims for realism and immersion, hypermediacy embraces artificiality and creates a distinctive visual style that draws attention to the medium-specific features of the film.

I examined the vast field of visual effects from different perspectives. Firstly, on the level of the fundamental component of film, the image or shot, which has undergone a significant ontological change. The key difference between a photochemical and a digital image is the inherent mutability of the digital that consists of pixels, easy to manipulate, change or distort, and all the operations are reversible. The plasticity of the digital image challenges classic realism theories while the indexicality of it is weakened, furthermore, the boundaries between real and artifice became blurred. Live-action cinematography, and the photographic image, which was the foundation of cinema for a century, became only an option and now can be easily manipulated and seamlessly blended with computer animation. As Manovich (2002) observed, as digital cinema opens to the painterly, “the history of the moving image makes a full circle. Born from animation […], only to become one particular case of animation in the end” (pp. 254-255).

Secondly, I examined the narrative integration of the spectacular visual effects from the viewer’s perspective. I derived from the example of *Jurassic Park (1993)* and through Gunning’s (1990), Mulvey’s (1975), and Whissel’s (2014) theories how to achieve the diegetic absorption of the spectacle without disrupting the audience's suspension of disbelief.

Thirdly, on an aesthetic level, I found a paradox: however digital imagery created almost limitless possibilities, we are still bound to the aesthetic of photorealism, and visual conventions of traditional film remained. Classic Hollywood narration’s desire for transparency and invisibility aims to keep the manipulation and the technology hidden, as well as the change of the medium. Digital imagery imitating film, resulting in a new kind of synthetic reality that Prince (1996) calls ‘perceptual realism’, that can be achieved, for instance, through ‘staged indexicality’ (Whissel, 2014). This involves imitating objects and their integration into Newtonian space without physical referents, resulting in a reality shaped by discourse rather than ontological indexicality. In other words, reality becomes a product of our perception rather than the inherent properties of objects in the world.

Finally, I examined the intricate relationship between editing and visual effects in practical terms. Both are key techniques of manipulation and creation of meaning based on the juxtaposition of different elements - editing involves the construction of images in time, while visual effects and composite involve intervention within the image. With digital technology, editors and filmmakers now have greater control over the image not only in time but in space as well. The relative effortlessness of making changes within the image allows using composite as an extension of editing, expanding the creative possibilities in solving narrative problems and making enhancements. These editorial fixes are part of the invisible visual effects that were not planned but came up during the editing process. Furthermore, in the case of visual effect-heavy films, the editor’s workflow becomes more complex, the involvement is longer, and new skills and approach are needed for assembling certain scenes with visual effects.

While the current mainstream visual effects workflows and leading technology have been discussed, it is important to note that creative and visual technology is constantly evolving. If the current digital technology and visual effects are challenging realism theories, the emergence of AI tools will further deepen the tension between reality and artifice, and once again, we will need to redefine our relationship to moving images and the creative industries. As digital technology continues to evolve and change the way we make and watch films, editors and filmmakers will continue to push the boundaries of what is possible while striving to maintain the artistry and realism that defines the medium.

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